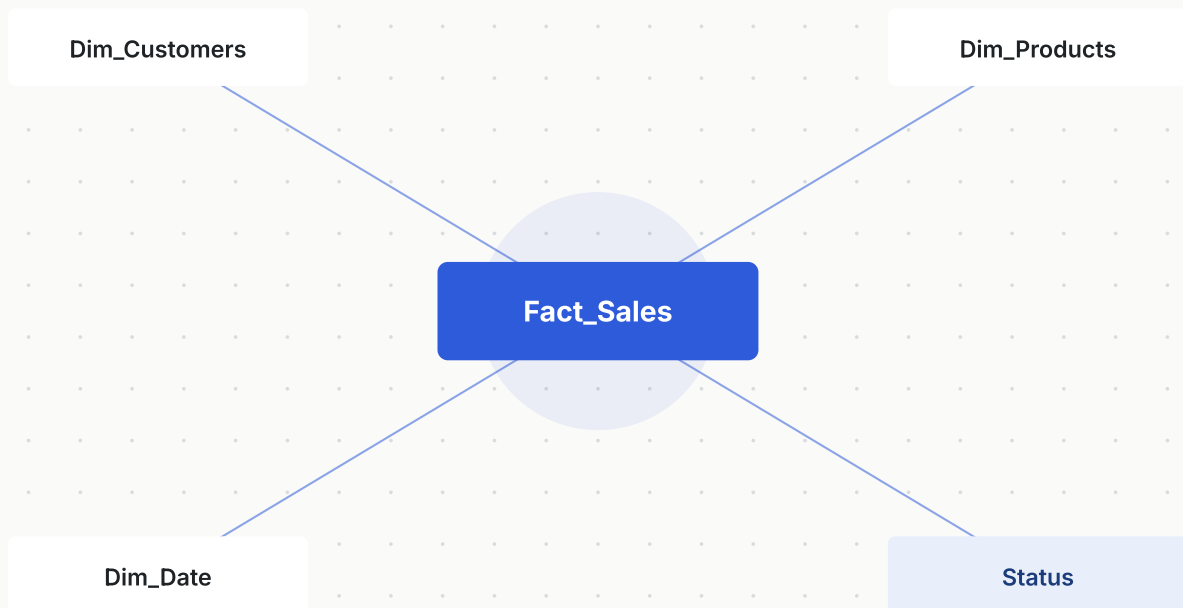




DATA ENGINEERING & ANALYTICS

North Meridian Retail Group.

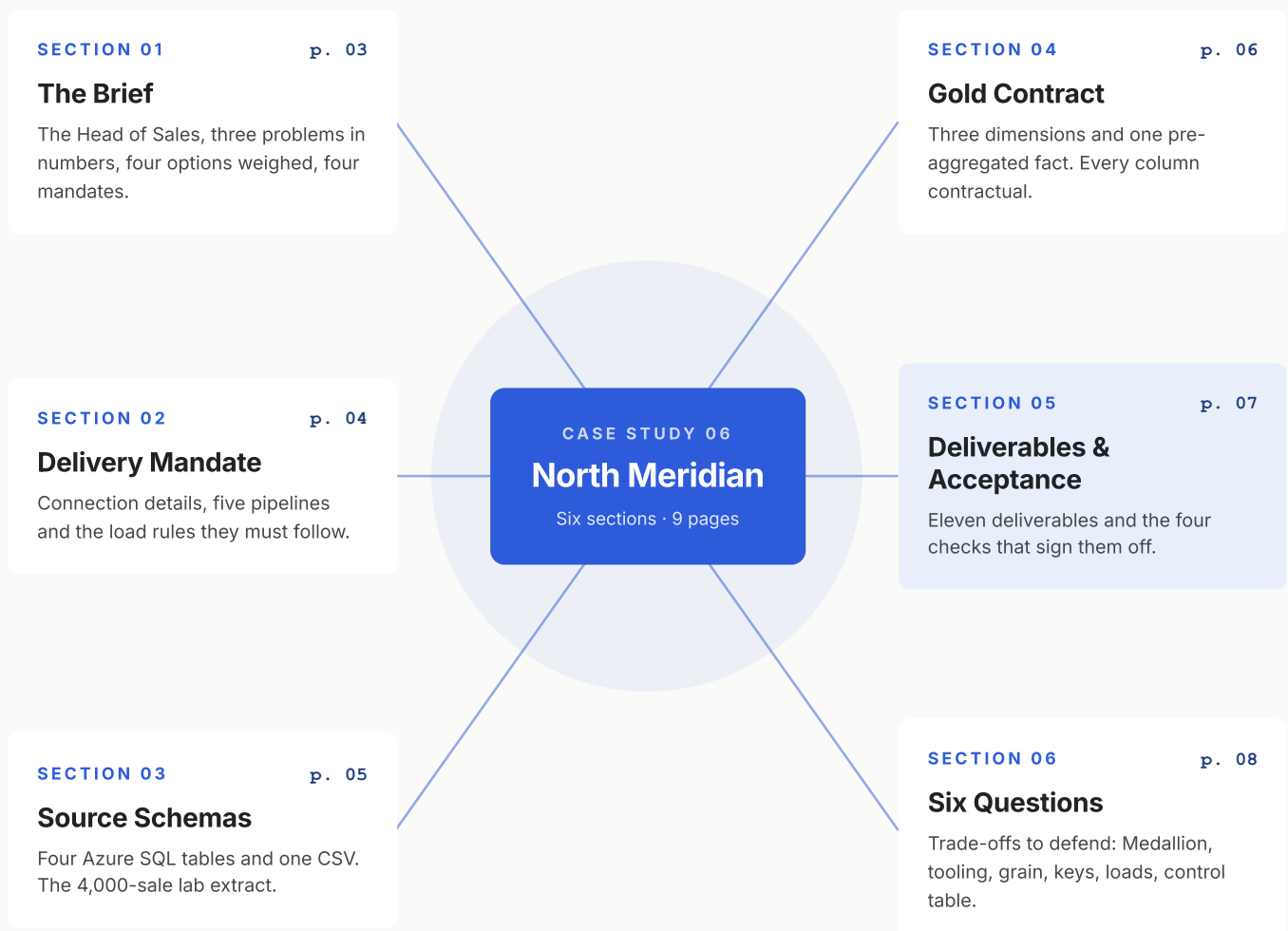
Unified Commerce Analytics: a Medallion lift from SQL to a Gold star schema.

A Microsoft Fabric case study in medallion architecture, Bronze-Silver-Gold transformation, star schema modelling, and Power BI semantic models for a multi-channel retail group escaping its legacy normalised warehouse.

CASE STUDY 06 · CONTENTS

One brief, six joins.

A retail group growing double digits online and stalling on reporting. A normalised SQL database that needs five joins for revenue by region. Thirty-six analyst hours a week lost to prep. A region mapping CSV that lives on someone's desktop. This is the brief.



Every section joins to the brief at the centre. Read 01 to 03 down the left, then 04 to 06 down the right.

SECTION 01 · THE BRIEF

Migrate to a Lakehouse. Without disrupting the till.

North Meridian sells online and in store, and online volume is growing double digits a year. Every report still runs against the Azure SQL database that serves the tills. Leadership wants analytics moved to a Fabric Lakehouse without touching the transaction flow.

"I am tired of hearing about database normalisation as an excuse for why I cannot see yesterday's sales. I do not care about Third Normal Form. I care about what my top ten products are right now."

HEAD OF SALES, NORTH MERIDIAN RETAIL GROUP

5

JOINS PER REVENUE QUERY

Normalised source

Revenue by region needs Sales, Orders, Products, Customers and the region CSV joined by hand. Business users cannot build their own reports.

3

COPIES OF THE REGION CSV

Shadow reference data

The state-to-region mapping lives in a CSV on analyst desktops. Three versions are in circulation, so regions report different totals.

36h

ANALYST PREP PER WEEK

Manual remediation

Three analysts spend about twelve hours a week each rewriting joins and patching fourteen saved SQL scripts that hardcode region logic.

FOUR OPTIONS, ONE RECOMMENDATION

Do nothing (baseline) REJECTED	FIXES Nothing to build.	LEAVES BROKEN Five-way joins on the tills' database, three CSV versions, 36 hours a week of prep.
Tune Azure SQL REJECTED	FIXES Faster queries through views and indexes.	LEAVES BROKEN Reports still load the tills. Region logic stays in fourteen scripts. The CSV stays on desktops.
Power BI on Azure SQL REJECTED	FIXES Dashboards within days.	LEAVES BROKEN Joins and region logic move into each report. Production still carries the load.
Fabric Lakehouse, star schema RECOMMENDED	FIXES Reporting off the tills. One region mapping. Revenue defined once. Full replay from source.	LEAVES BROKEN Build effort. Data refreshes daily at 06:00, not live.

Only the Lakehouse takes reporting off the tills and puts the logic in one place. The price is a daily refresh, and the Head of Sales accepts it: he needs yesterday's sales by 08:00, not live ones.

FOUR MANDATES

01 Replayability

Full historical reloads from source without duplicating records.

Met by: [Bronze, Pipelines 1 and 2](#)

03 SLA observability

Yesterday's sales land in Gold before the 08:00 reporting window, every day.

Met by: [Master Pipeline, 06:00 trigger](#)

02 Schema visibility

Every step from source to star schema lives in version-controlled notebooks.

Met by: [Deliverables 6 and 7](#)

04 Domain ownership

The Sales team owns the revenue definition; the platform owns the pipes.

Met by: [Fact_Sales, Semantic Model](#)

SECTION 02 · DELIVERY MANDATE

Four pipelines. Two notebooks. One master.

You are the Lead Data Engineer. Build the platform from the Azure SQL source to a Power BI star schema. RegionMapping.csv is uploaded to the Lakehouse Files area.

EXHIBIT A · AZURE SQL CONNECTION

SERVER

fabric-project-srv-2027.database.windows.net

Fully qualified Azure SQL server hostname. [Set in: Linked Service](#)

SCHEMA

NorthMeridianRetail

All four tables live here, not in dbo. [Set in: Source query](#)

DATABASE

msfabricbootcamp

Target database name. [Set in: Linked Service](#)

AUTHENTICATION

SQL Authentication

Credentials from the instructor. Store as a Fabric Secret, not plaintext. [Set in: Linked Service](#)

EXHIBIT B · PIPELINES

RUN ONCE, BY HAND

Pipeline 1 - Sales Batch

BATCHED FULL LOAD

Sales in SalesID slices driven by ControlTable.

Pipeline 2 - Orders Batch

BATCHED FULL LOAD

Orders in OrderID slices driven by ControlTable.

DAILY AT 06:00 · MASTER PIPELINE · EACH STEP WAITS FOR THE LAST

Pipeline 3 - Incremental

INCREMENTAL

Sales and Orders rows added since the last successful run.



Pipeline 4 - Dimensions

FULL REFRESH

Customers and Products, replaced in full.



Bronze to Silver

NOTEBOOK

Silver tables as Delta.



Silver to Gold

NOTEBOOK

Gold star schema.



Semantic Model

REFRESH

Power BI reads the new Gold.

EXHIBIT C · LOAD RULES

How you meet these rules is part of the lab.

01

Batched full loads

4 SLICES EACH · PIPELINES 1 AND 2

Sales (SalesID 50001–54000) and Orders (OrderID 70000–73999) each load in four slices of 1,000, one row per slice in ControlTable. A failed slice reruns without reloading the others.

02

New rows only

INCREMENTAL · PIPELINE 3

Each daily run loads only rows added since the last successful run. The first daily run must not re-ingest history loaded by Pipelines 1 and 2.

03

Safe reruns

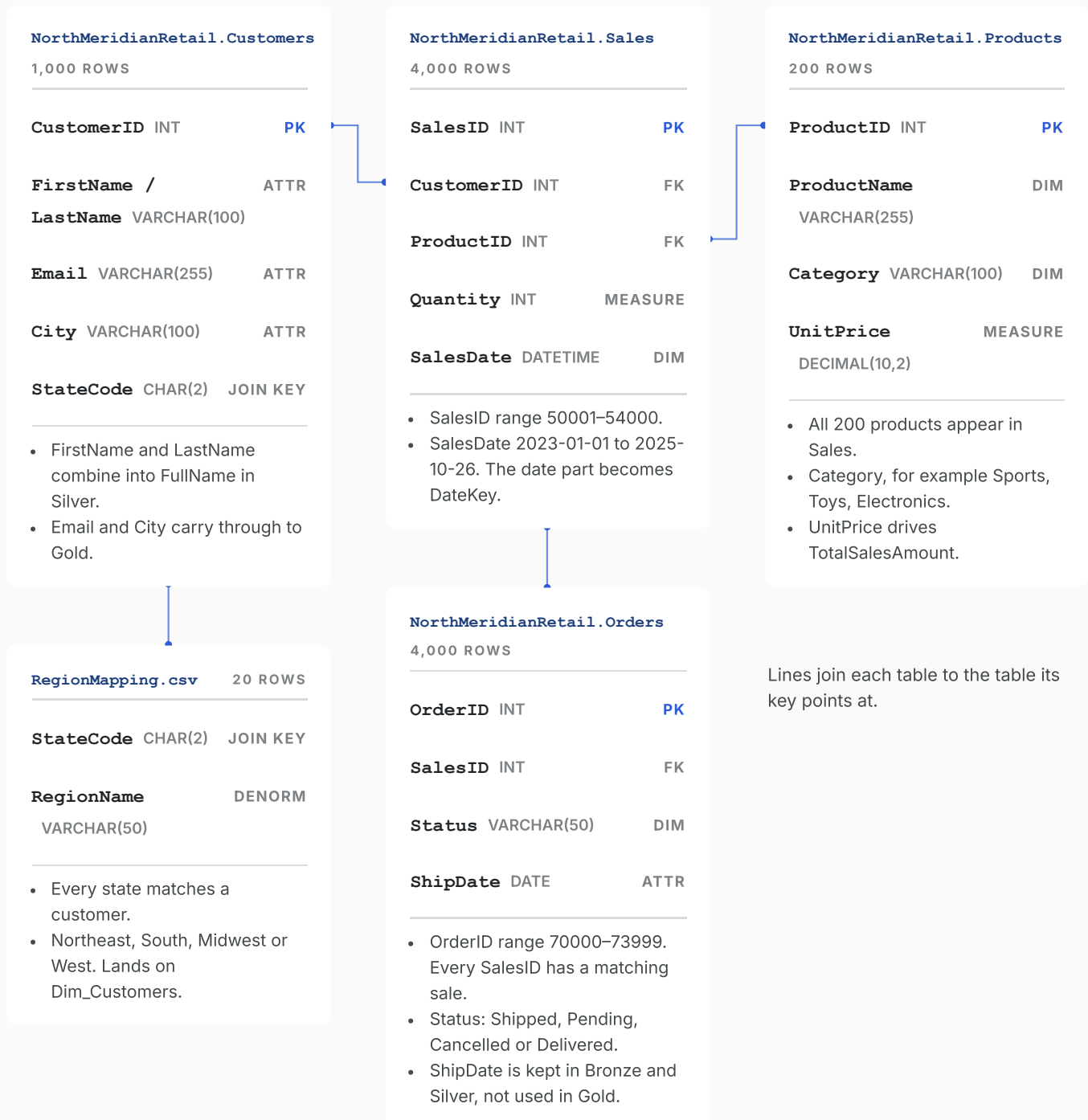
IDEMPOTENT · ALL

Re-running any pipeline or notebook must not create duplicate rows in any layer.

SECTION 03 · SOURCE SCHEMAS

Four Azure tables. One CSV.

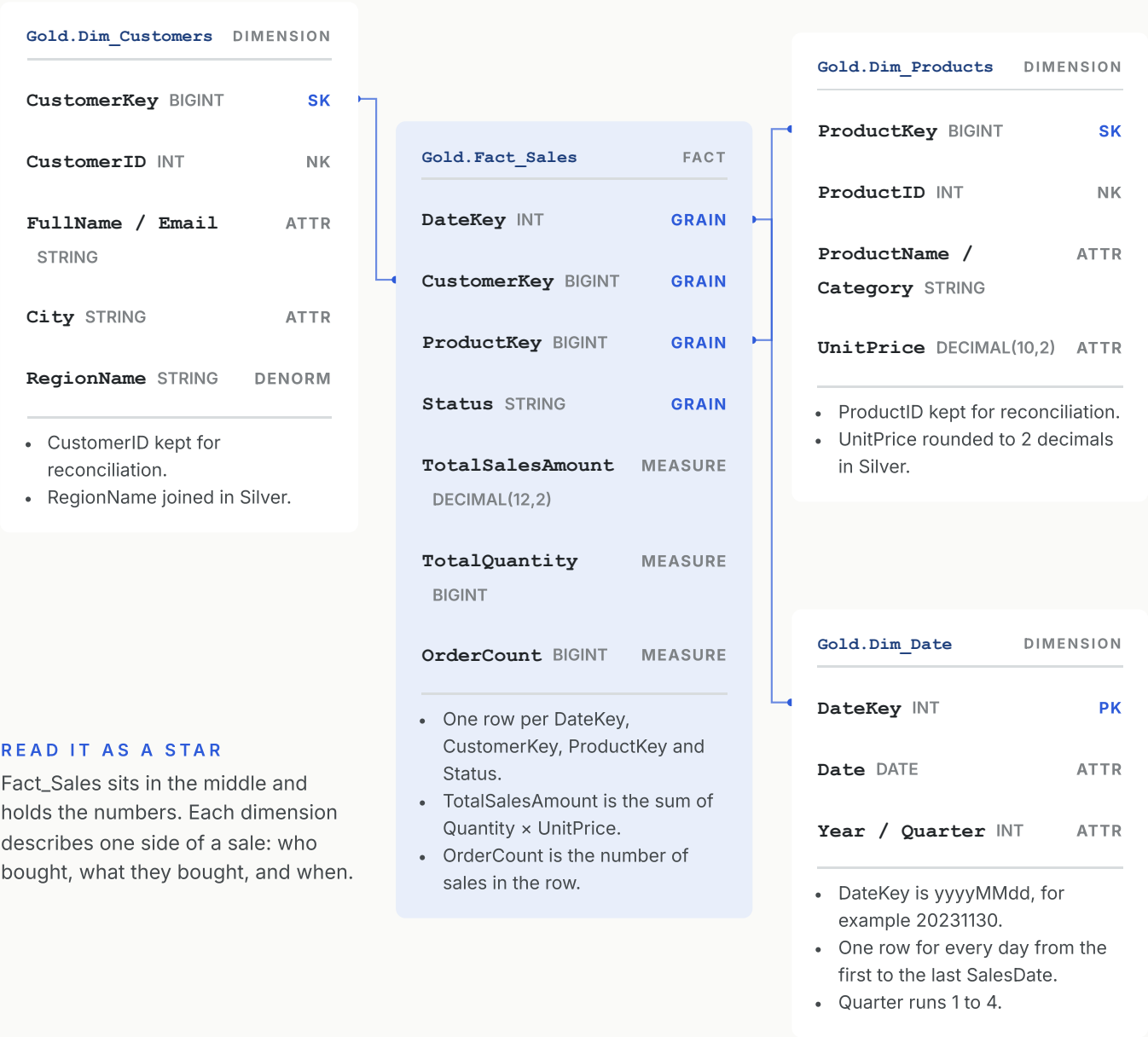
Bronze keeps every column as-is. The lab extract holds 4,000 online sales, January 2023 to October 2025.



SECTION 04 · GOLD CONTRACT

Three dimensions. One pre-aggregated fact.

Column names, types and grain are contractual. Every dimension joins Fact_Sales on its key.



SECTION 05 · DELIVERABLES & ACCEPTANCE

Eleven deliverables. Four checks.

Pipelines 1 and 2 and the control table run once, by hand. The daily chain runs under the Master Pipeline at 06:00.

BUILD AND LOAD ONCE

- 01

ControlTable

CONTROL

One row per Sales or Orders slice, with a way to skip slices already loaded.
- 03

Pipeline 2 - Orders Batch

ONE-TIME

Orders to Bronze one slice at a time from ControlTable. Run once.

- 02

Pipeline 1 - Sales Batch

ONE-TIME

Sales to Bronze one slice at a time from ControlTable. Run once.

DAILY CHAIN AND REPORTING

- 04

Pipeline 3 - Incremental

DAILY

New Sales and Orders rows since the last successful run.
- 06

Bronze to Silver Notebook

3 TABLES

Silver_Customers, Silver_Products and Silver_Sales as Delta.
- 08

Semantic Model

MODEL

Direct Lake on Gold. Fact_Sales to each dimension on its surrogate key, one-to-many.
- 10

Master Pipeline

CHAIN

Pipelines 3 and 4, both notebooks, then Semantic Model refresh.

- 05

Pipeline 4 - Dimensions

DAILY

Customers and Products refreshed in full.
- 07

Silver to Gold Notebook

STAR

The four Gold tables, exactly as the Gold contract defines them.
- 09

Power BI Report

REPORT

Line chart of TotalSalesAmount by Date. Bar chart of the top 10 products.
- 11

Scheduled Trigger

06:00

Daily at 06:00 on the Master Pipeline. Pipelines 1 and 2 are not scheduled.

ACCEPTANCE CHECKS

G1

Completeness.

A SQL Endpoint query shows Bronze row counts equal source for every table. OrderCount in Fact_Sales sums to the Silver_Sales row count.

G2

To the cent.

SUM(TotalSalesAmount) in Gold equals SUM(Quantity × UnitPrice) in Bronze. Variance allowed: \$0.00.

G3

Incremental.

A second daily run with no new source rows adds nothing to any layer.

G4

Freshness.

The Master Pipeline finishes within thirty minutes of the 06:00 trigger.

SECTION 06 · DISCUSSION

Six questions. Trade-offs you must defend.

Each question targets a decision you must justify to a technical or executive audience.

Q1 Medallion rationale

Why land raw data in Bronze instead of writing straight to Silver? Which failure can you only recover from because Bronze exists? Tie your answer to the Replayability mandate.

Q2 Notebook vs Dataflow Gen2

Silver needs a three-way join (Sales, Orders, Products), a CSV join and derived columns. Would you build it in a PySpark Notebook or Dataflow Gen2, and what constraint decides it?

Q3 Fact_Sales grain

Fact_Sales is pre-aggregated by DateKey, CustomerKey, ProductKey and Status. What does this grain do for Power BI performance under VertiPaq's columnar compression, and what does an analyst lose?

Q4 Surrogate keys

Which PySpark function generates the surrogate keys in this lab? What is its limitation on a distributed cluster, and what must change so the keys stay stable across daily loads?

Q5 Batch vs incremental

Why do the initial loads use range slices while the daily load uses an ID watermark? Why is an ID watermark safer here than a LastModifiedDate column?

Q6 Control table

A partial Orders load fails halfway through. What does the control table let you do on the rerun that one large Copy cannot?

From five-way SQL joins to a star schema the board can read.

Build the Medallion lift that gives North Meridian a dashboard worth opening at 08:00. Bronze that replays without duplication. Silver that joins Sales, Orders, Products, Customers and RegionMapping into one analytical contract. A Gold star schema with surrogate keys and pre-aggregated facts. One scheduled pipeline. One sum that reconciles to the cent.

SIGN-OFF SHEET · ELEVEN DELIVERABLES

☐ 01 ControlTable	☐ 02 Pipeline 1 - Sales Batch	☐ 03 Pipeline 2 - Orders Batch
☐ 04 Pipeline 3 - Incremental	☐ 05 Pipeline 4 - Dimensions	☐ 06 Bronze to Silver Notebook
☐ 07 Silver to Gold Notebook	☐ 08 Semantic Model	☐ 09 Power BI Report
☐ 10 Master Pipeline	☐ 11 Scheduled Trigger	

HOW YOU WILL BE EVALUATED

The four acceptance checks on page 07: row counts, the sum to the cent, a second incremental run that adds nothing, and a Master Pipeline inside thirty minutes. Six discussion answers framed against the mandates.

WHAT THIS PROVES

You can move analytics off a production database without disrupting the tills. You can build the Medallion as three tiers of trust, not three folders. You can explain what keeps a surrogate key stable across incremental loads.